

ELITE FOUR CAPITAL

Political Event & Relative-Value Desk

Pricing, Arbitrage, Fees, and Timing in Binary Political Prediction Markets

A quantitative framework, applied to the 2028 Jon Ossoff complex

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1. Executive summary

A binary prediction-market contract pays \$1 if the event occurs and \$0 if it does not. Therefore its price is a tradable estimate of the event's probability. This paper defines how the desk prices, cross-checks, sizes, and executes these contracts. It shows where fees, venue choice, and settlement timing change the answer. The worked example is the set of markets on Senator Jon Ossoff for 2028. All figures are live as of 16 July 2026.

The desk finds five edges. They run from cleanest to least clean:

- A hard no-arbitrage constraint. A win in the presidency requires the nomination. Therefore $P(\text{presidency})$ is always less than or equal to $P(\text{nomination})$. A monitor flags any violation as a locked arbitrage or a data error.
- A cross-venue lock. The same contract has a different price on Polymarket and Kalshi. Buy the Yes on the cheaper venue and the No on the richer venue. The pair pays \$1 in every outcome for less than \$1. This is a true arbitrage, after fees and the cost of two platforms.
- A Dutch book on the overround. The Kalshi candidate field sums to about 105 percent. Buy the No on every name to lock about 5 percent, minus fees. Polymarket links its field to about 100 percent, so it has no Dutch book.
- A relative-value wedge. The two markets imply a conditional general-election win of about 52 to 60 percent if Ossoff is nominated. This is rich for a first-term senator. Long the nomination and short the presidency to isolate the nominated-but-loses outcome cheaply.
- A timing edge. The primary calendar decides the nomination. The nomination contract re-rates to about 99 cents months before the November 2028 settlement. Harvest the move; do not hold to term.

Note three caveats. First, all the markets load on one factor: Ossoff's standing. Second, the edges are perishable; the Ossoff cross-venue gap reversed within a day on 16 July. Third, venue choice trades price against fees, tax cleanliness, and counterparty safety.

2. Pricing framework

For an event E , a \$1-on-Yes contract that trades at q between 0 and 1 satisfies $q = P(E)$, the risk-neutral probability, when there are no fees and no time value. Call the desk's fair estimate p . The signed edge on a long Yes is p minus q . The live inputs are: nomination q_N about 0.14, presidency q_P about 0.075, and horizon τ about 2.31 years.

3. The no-arbitrage subset constraint

Ossoff reaches the presidency only as the nominee, so P is a subset of N . Therefore:

$$q_P \leq q_N.$$

If this fails, buy the Yes-nomination and the No-presidency. The pair returns at least \$1 in every outcome, so any cost below \$1 is riskless. The desk monitors this constantly. It is both a mispricing alarm and a data-integrity check; a stale feed once inverted it.

4. Conditional decomposition and the wedge

By definition $q_P = q_N$ times c , so the market's implied conditional is $c = q_P / q_N$, about 0.52 to 0.60. This is demanding for a first-term swing-state senator; a generic out-party nominee is about 0.50, and a lesser-known candidate runs behind. The wedge is long the nomination and short the presidency. It isolates the set N minus P , the nominated-but-loses outcome. The market prices it at $w = q_N$ minus q_P , about 5 cents,

against a modelled fair of p_N times $(1 - \hat{c})$, about 6 to 8 cents. Its only residual exposure is to c , and it is hedged against Ossoff's overall trajectory.

View sensitivity. If your estimate of c is above the market's, the wedge is the wrong trade. For example, you may believe that a strong nominee rides an anti-incumbent wave to the White House. In that case, buy the presidency directly at about 7 cents for the largest multiple.

5. Cross-venue pricing: overround and fees

The same nomination event trades richer on Kalshi than on Polymarket. Two effects cause this:

- **Overround.** Kalshi lists each candidate as an independent market, so the field sums to about 105 percent, a 5-point house margin. Every Yes runs rich. Polymarket's neg-risk linkage arbitrages the field to about 100 percent. On 16 July, Ossoff was 12 cents on Polymarket against 15 cents on Kalshi, and Beshear was 2.1 cents against 3.8 cents.
- **Clientele lean.** On two-way partisan markets the gap flips with the crowd. In 2024, Polymarket ran 5 to 8 points hotter for Trump than Kalshi.

Fees. Both venues charge a taker fee of the form $\text{coeff} \times \text{price} \times (1 - \text{price})$. Kalshi's coefficient is 0.07; Polymarket's is 0.04. Makers pay zero. As a share of the stake, the fee is $\text{coeff} \times (1 - \text{price})$, so it is worst on a cheap long shot.

Contract price	Kalshi fee (of stake)	Polymarket fee (of stake)
3¢ (deep longshot)	6.8%	3.9%
14¢ (Ossoff)	6.0%	3.4%
50¢ (toss-up)	3.5%	2.0%
90¢ (favorite)	0.7%	0.4%

All-in cost. Combine price and fee into an all-in cost of $q \times (1 + \text{coeff} \times (1 - q))$. Polymarket usually wins for buying the Yes on a candidate long shot, but the edge is perishable. On 16 July, Ossoff's Polymarket price rose from 12 cents to 14.4 cents, and the best venue flipped from Polymarket (all-in 12.4 cents) to Kalshi (14.8 cents) within the hour.

6. The arbitrage scan

Cross-venue lock. Buy the Yes on the cheaper venue and the No on the richer venue. One leg pays \$1 in every outcome, at a cost of $q_{\text{cheap}} + (1 - q_{\text{rich}})$. This locks $q_{\text{rich}} - q_{\text{cheap}}$, minus the two taker fees. The payoff is a true arbitrage. The frictions are separate: collateral on two platforms, fees, US-access rules, and the need for both venues to resolve identically on the same event and source.

Dutch book on the overround. With the Kalshi field at about 105 percent, buy the No on all N candidates. The cost is the sum of $(1 - q_i)$, which equals $N - 1.05$. The book pays \$1 on the $N - 1$ losers, for about 5 percent gross, minus fees. The risks are an incomplete fill across illiquid names and a write-in or late entrant. Polymarket's field at about 100 percent offers no such book.

The dashboard shows these on the Arb tab. All are net of fees. The dashboard flags one only when the edge clears the frictions.

7. Settlement timing — the convergence clock

The nomination contract settles in November 2028, but the primary calendar (February to June 2028) decides it. It converges to about 99 cents by the August convention. The desk harvests the re-rating and exits early,

rather than holding to term. This collapses a 2.3-year lock into weeks. The presidency contract does not converge early; it stays live to Election Night. A withdrawal announcement collapses the nomination Yes to a few cents in one step.

8. Position sizing (Kelly)

For a binary at price q with fair value p , the net odds are $b = (1 - q) / q$, and the growth-optimal fraction is $f^* = (p - q) / (1 - q)$. At $q = 0.14$ and $p = 0.15$, f^* is about 1.2 percent of bankroll at full Kelly. The desk sizes at quarter-Kelly. The three markets are highly correlated, so the desk reduces the effective fraction further at the book level.

9. Time value and the hold-versus-trade decision

Collateral earns no yield, yet resolution is about 2.3 years out. The expected annualised return if held is (p / q) to the power $(1 / \tau)$, minus 1, which is about 6 to 7 percent. This is only about 2 to 3 percent above cash, and it is attached to a lumpy binary payoff, so it is unattractive on a risk-adjusted basis. With Section 7, the conclusion is to trade the re-rating, not hold to resolution. Polymarket's continuous order book makes an early exit routine.

10. Regulation, counterparty, and payout

Dimension	Kalshi	Polymarket
Counterparty	CFTC-regulated clearinghouse (Kalshi Klear) guarantees payout	Smart contract + crowd; an oracle (UMA) resolves
Your funds	Segregated, bankruptcy-remote USD (JPMorgan, BNY Mellon)	Self-custodied USDC; no segregation, no insurance
Shutdown risk	Lower — federally regulated; won its CFTC fight	Mixed — geo-block history + state actions (MN, NV)
Tax record	USD; limited 1099s; cleaner	USDC = every trade also a crypto disposal

Neither venue is FDIC-insured or SIPC-insured. The Kalshi price premium of about 2 to 3 cents is, in effect, a safety-and-tax cost. You pay more for a guaranteed payout and cleaner records. Whether it is worth paying depends on your size and holding period.

11. Risk model and recommended positions

- **Single-factor concentration.** All the markets load on Ossoff's standing. Treat them as one factor, and cap the total at 5 percent of bankroll or less.
- **Perishability.** The edges converge fast, as the 16 July reversal showed. Enter on a dislocation and exit on the re-rating.
- **Data quality.** Use the per-market endpoints with timestamps. The aggregate cache was found 30 days stale, so the desk does not trust it for execution.

Trade	Rationale	Sizing
Long nomination (cheaper venue)	Overround + fee edge on Polymarket; harvest the primary re-rating.	¼-Kelly
Wedge: long Nom / short Pres	Fades a rich ~52–60% conditional; hedged to Ossoff's path.	Primary RV
Cross-venue lock / Dutch book	When the Arb tab shows a net-of-fee edge and both venues are accessible.	Opportunistic

Trade	Rationale	Sizing
Buy presidency (only if bullish)	For a “wins it all” view — biggest multiple, cheapest on Polymarket.	Directional

Appendix — key formulas & sources

- Price = probability; edge = p minus q . No-arbitrage: $q_P \leq q_N$. Conditional $c = q_P / q_N$.
- Wedge price = q_N minus q_P ; fair = p_N times (1 minus c). Kelly $f^* = (p \text{ minus } q) / (1 \text{ minus } q)$, times one-quarter.
- Taker fee = coeff times q times (1 minus q); coeff is 0.07 (Kalshi) or 0.04 (Polymarket); all-in = q times (1 + coeff times (1 minus q)).
- Cross-venue lock = q_{rich} minus q_{cheap} , minus fees. Dutch book = the sum of (1 minus q_i) against an (N minus 1) payout = overround minus fees.
- Annualised IRR if held = (p / q) to the power $(1 / \tau)$, minus 1. Time-value hurdle = $(1 + r_f)$ to the power τ .

Sources: Polymarket gamma API and Kalshi trade API (live, 16 July 2026); Kalshi and Polymarket fee schedules; Sherwood/CFTC on 2024 cross-venue divergence. Educational research; not investment advice.